

Jiayang Song

ASSISTANT PROFESSOR · LEIDEN UNIVERSITY

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Research Interests

- Trustworthy AI Systems
- Cyber-Physical Systems
- Foundation Model
- Software Engineering
- Quality Assurance
- Embodied Agent

Work Experience

Assistant Professor

Leiden University, Netherlands

LEIDEN INSTITUTE OF ADVANCED COMPUTER SCIENCE (LIACS)

10.2026 - Current

- Trustworthy AI System Engineering
- Multimodal Foundation Model
- Software and AI in Business Cluster

Assistant Professor

Macau University of Science and Technology, Macao

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

09.2025 - 08.2026

- Software Engineering for AI Systems

Graduate Research Assistant Fellowship

University of Alberta, Canada

ADVISORS: PROF. LEI MA

01.2022 - 05.2025

- Quality and Safety Assurance for AI System Engineering
- University research projects/grants: FES, MIF-RCES, Amii-RAP
- Industrial collaboration projects: Nvidia, TIER IV

Education

University of Alberta

Edmonton, Canada

PH.D., ELECTRICAL AND COMPUTER ENGINEERING

01.2022 - 05.2025

- Area of Interest: Software Engineering and Intelligent System
- Thesis: Quality Assurance for Trustworthy AI-enabled Cyber-Physical System (AI-CPS)
- Advisor: Prof. Lei Ma

University of Toronto

Toronto, Canada

M.ENG, ELECTRICAL AND COMPUTER ENGINEERING

09.2019 - 06.2021

- Graduate with distinction
- Specialization: Deep Reinforcement Learning, Data Science

Western University

London, Canada

B.ENG, ELECTRICAL AND COMPUTER ENGINEERING

09.2015 - 04.2019

- Graduate with distinction
- Specialization: Control System, Wireless Communication
- Dean's Honor List 2017, 2018, 2019

Research Project and Collaboration

Uncertainty Estimation and Confidence Calibration for Multimodal LLMs

MUST

PRINCIPAL INVESTIGATOR

2026 - Current

- Capability Enhancement Grant from Science and Technology Development Fund (FDCT)
- Developing a framework to calibrate MLLM's confidence during generation

Testing and Evaluation Framework for Generative AI-enabled Embodied Systems

MUST
2025 - Current

PRINCIPAL INVESTIGATOR

- General Research Grant from MUST
- Developing testing frameworks for embodied AI in robotics manipulation

Automatic Reward Design and Optimization for Reinforcement Learning in Robotics

MUST
2026 - Current

PRINCIPAL INVESTIGATOR

- Overseas Research Fund from State Key Lab of Novel Software Technology
- Iterative Reward Design and Refinement empowered by Multimodal AI with Real2Sim Feedback.

Foundational Models for Autonomous Driving System

University of Alberta
2024 - Current

Co-PI

- Collaboration with TIER IV, Tokyo, Japan
- Exploring the potential of Multimodal Foundation Model-driven autonomous driving systems
- Developing novel frameworks for ADS scenario understanding, annotation and prediction

Model-based Analysis and Testing Guidance for Autonomous Driving System

University of Alberta
2023 - 2024

RESEARCH ASSISTANT

- Industry collaboration
- Developed a model-based ADS test case selection framework
- The designed framework has been applied and validated on real ADS development

Safety and Reliability Assurance of Next Generation AI-CPS in Energy Systems

University of Alberta
2022 - 2024

HIGH QUALITY PERSONNEL (HQP)

- University research project: Future Energy Systems (FES)
- Developed safety enhancement and monitoring frameworks for AI-CPS across various application domains
- Designed two automated repair techniques for AI controllers

Benchmarking and Evaluating AI-CPS for Robotic Manipulation

University of Alberta
2022 - 2023

RESEARCH ASSISTANT

- Collaboration with Nvidia AI Tech Centre, Singapore
- Developed a benchmark and a testing framework for AI-CPS in robotic manipulation using NVIDIA Isaac Sim

Teaching Experience

Software Engineering and Project Management

MUST
2026

COURSE INSTRUCTOR

- Bachelor's course (approx. 120 participants per semester)
- Software development foundation and quality assurance techniques

Programming Language

MUST
2025 - 2026

COURSE INSTRUCTOR

- Bachelor's course (approx. 100 participants per semester)
- C language foundation

Exploring Software Development Domains

University of Alberta
2023 - 2025

TEACHING ASSISTANT

- Bachelor's course (approx. 60 participants per semester)
- Advanced software engineering concepts using Rust

Introduction to Digital Logic Design

University of Alberta
2023 - 2024

TEACHING ASSISTANT

- Bachelor's course (approx. 250 participants per semester)
- Introduction to computer-aided design and simulation tools for digital circuit design

Fundamentals of Electrical Engineering

University of Alberta

TEACHING ASSISTANT

2023 - 2024

- Bachelor's course (approx. 300 participants per semester)
- Physical concepts of passive circuit elements, Kirchhoff's laws and DC circuit equations

Analog Electronics

University of Alberta

TEACHING ASSISTANT

2023

- Bachelor's course (approx. 300 participants per semester)
- Circuit design with feedback topologies and amplifiers

Professional Activities

TALKS

- **Invited Talk at APSEC 2025**
 - Topic: A New Journey: Navigating the Cold Start in First-Year Learning
- **Invited Talk at University of Alberta as Guest Lecture, Edmonton, Canada (2023 - 2025)**
 - Topic: Quality Assurance for AI-enabled Cyber-Physical Systems
- **Invited Talk at East China Normal University, Shanghai, China (2023)**
 - Topic: AI-enabled Cyber-Physical Systems and Software Foundation

PC MEMBER

- ACM International Conference on the Foundations of Software Engineering (FSE 2027)
- International Workshop on Infrastructure for Trustworthy Software Agents (AGENTVERIFY 2027)
- Conference on Software Analysis, Evolution and Reengineering (SANER 2026)
- ACM International Conference on the Foundations of Software Engineering (FSE 2026): IVR Track
- IEEE Annual International Computer Software and Applications Conference (COMPSAC 2026): SETA
- CCF International Conference on Service Science (ICSS 2026)

REVIEWER

- IEEE Transactions on Software Engineering (TSE)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- Empirical Software Engineering (EMSE)
- Automated Software Engineering (AUSE)
- IEEE Transactions on Reliability (ToR)